

5 Theoretical model

The system is described by one Lagrangian: the kinetic and elastic energies of two vibrating cantilevers plus the interaction energy of two magnetic dipoles. Everything measured in this essay — the two mode frequencies, their dependence on d , the peak heights, the beat minima, the decay — follows from it. Sections 5.1–5.6 develop the linear theory that the main analysis tests; Section 6 (extension) keeps the full force law and predicts how the linear results fail as the magnets get close.

5.1 Energy of one vibrating leaf spring

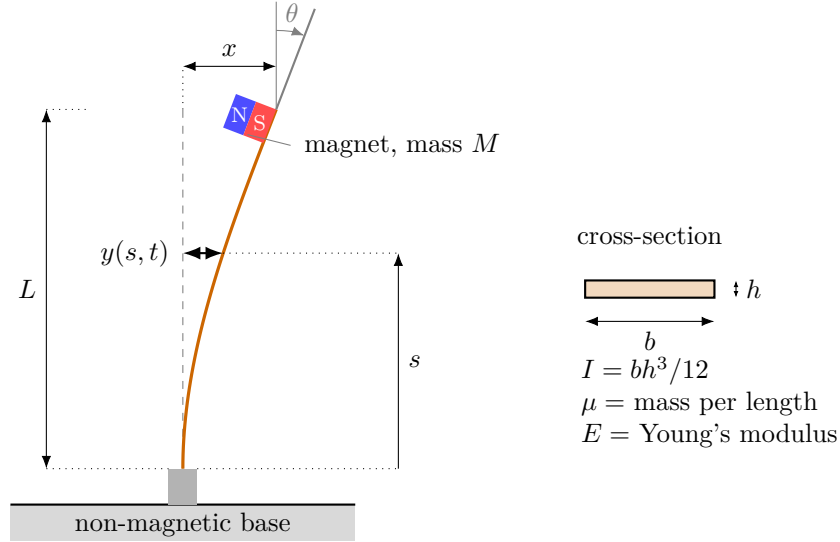


Figure 1: One leaf spring: free length L above the clamp, lateral deflection $y(s, t)$, tip displacement $x(t) = y(L, t)$, tip rotation θ , magnet of mass M at the tip.

Separation of variables. The strip is a thin elastic beam (Euler–Bernoulli: plane sections stay plane, deflections small). The lowest vibration mode of a cantilever with a heavy tip mass is, to very good accuracy, the *static* deflection curve produced by a tip force [1, 2]:

$$y(s, t) = x(t) \varphi(s), \quad \varphi(s) = \frac{3Ls^2 - s^3}{2L^3}, \quad \varphi(0) = \varphi'(0) = 0, \quad \varphi(L) = 1, \quad \varphi''(L) = 0. \quad (1)$$

The shape satisfies the clamped conditions at $s = 0$ and zero bending moment at the free tip. Only the tip coordinate $x(t)$ is then a dynamical variable — the beam has been reduced to one degree of freedom (Rayleigh’s method).

Kinetic and elastic energy. Every element μds of the strip moves with velocity $\dot{x}\varphi(s)$; the magnet moves with $\dot{x}$. With bending energy $\frac{1}{2}EI \int (\varphi'')^2 ds$ (strain energy of a bent beam, [1]):

$$T_{\text{beam}} = \frac{1}{2}\mu\dot{x}^2 \int_0^L \varphi^2 ds = \frac{1}{2} \left(\frac{33}{140}\mu L \right) \dot{x}^2, \quad T_{\text{magnet}} = \frac{1}{2}M\dot{x}^2, \quad (2)$$

$$V_{\text{beam}} = \frac{1}{2}EIx^2 \int_0^L (\varphi'')^2 ds = \frac{1}{2} \left(\frac{3EI}{L^3} \right) x^2 \equiv \frac{1}{2}kx^2, \quad k = \frac{Ebh^3}{4L^3}. \quad (3)$$

(The integrals: $\int_0^L \varphi^2 ds = \frac{33}{140}L$ and $\int_0^L \varphi''^2 ds = 3/L^3$.) So one spring with its magnet is a harmonic oscillator with

$$m = M + \frac{33}{140}\mu L, \quad k = \frac{Ebh^3}{4L^3}, \quad \omega_0^2 = \frac{k}{m}. \quad (4)$$

Lagrange's equations $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_i} = \frac{\partial \mathcal{L}}{\partial x_i}$ give

$$m\ddot{x}_1 = -kx_1 - \frac{\partial U}{\partial x_1}, \quad m\ddot{x}_2 = -kx_2 - \frac{\partial U}{\partial x_2}. \quad (9)$$

For untilted moments $-\partial U/\partial x_1 = -F(r)$ and $-\partial U/\partial x_2 = +F(r)$ with r the current separation: the repulsion pushes magnet 1 left and magnet 2 right. Equations (9) are exact within the dipole and one-mode approximations; they are non-linear because $F \propto r^{-4}$, and Section 6 solves them as they stand. The main analysis uses their linearisation.

Equilibrium. At rest $kx_1 = -F(d)$, $kx_2 = +F(d)$: each spring is bent outwards by $\delta = F(d)/k$ and the magnets settle at $d = d_0 + 2\delta$ (Figure 2). From now on x_1, x_2 denote displacements from *these* positions; the constant forces cancel.

Small oscillations. Expanding U to second order in x_1, x_2 (and in the tilt angles) about equilibrium,

$$U \simeq U_0 + F(d)(x_1 - x_2) + \frac{1}{2}\gamma(x_2 - x_1)^2 - \frac{\gamma d^2}{128L^2}[9(x_1 + x_2)^2 + 3(x_2 - x_1)^2] + \dots, \quad (10)$$

$$\gamma \equiv -\left.\frac{dF}{dr}\right|_d = \frac{4F(d)}{d} = \frac{6\mu_0\mu_m^2}{\pi d^5}.$$

The first-order term is the equilibrium force already balanced by $k\delta$. The term $\frac{1}{2}\gamma(x_2 - x_1)^2$ is the energy of a spring of stiffness γ — the *magnetic stiffness* — joining the two magnets: repulsion that weakens with distance behaves, for small changes of separation, like a compressed spring. The last term is the tilt correction of (6): it lowers ω_1^2 and ω_2^2 by the relative amounts $\frac{9}{32}\frac{\gamma}{k}(d/L)^2$ and $\frac{3}{32}g(d/L)^2$ ($g = \gamma/(k + 2\gamma)$), i.e. by less than 10^{-3} for $d \leq 42$ mm and $L \geq 150$ mm — below the FFT resolution — and is dropped (Section 6 lists it among the neglected effects). The linear equations of motion are therefore

$$\boxed{m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1)} \quad (11)$$

— two equal masses on equal springs k , coupled by a spring γ .

5.4 Normal modes

The symmetry of (11) under $1 \leftrightarrow 2$ suggests the coordinates

$$\eta = x_1 + x_2 \quad (\text{centre of mass}), \quad \xi = x_2 - x_1 \quad (\text{change of separation}). \quad (12)$$

Adding and subtracting the two equations decouples them:

$$m\ddot{\eta} = -k\eta, \quad m\ddot{\xi} = -(k + 2\gamma)\xi \quad \implies \quad \boxed{\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k + 2\gamma}{m}} \quad (13)$$

In the *in-phase* mode ($\xi = 0$) the separation never changes, the magnetic force stays at its equilibrium value and each spring swings as if alone at ω_1 , whatever d is. In the *anti-phase* mode ($\eta = 0$) a displacement a of each magnet changes the separation by $2a$, adding a force $2\gamma a$ to ka : stiffness $k + 2\gamma$. In the Lagrangian picture, $\mathcal{L} = \frac{1}{4}m(\dot{\eta}^2 + \dot{\xi}^2) - \frac{1}{4}k\eta^2 - \frac{1}{4}(k + 2\gamma)\xi^2$ is a sum of two independent oscillators.

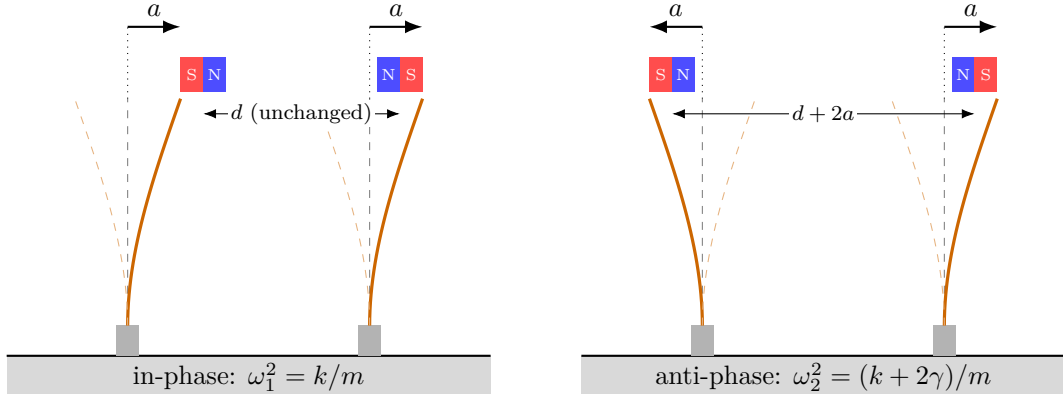


Figure 3: The two normal modes (solid: one extreme of the swing; dashed: the other; magnets drawn at the solid extreme). Left: both magnets move together, the separation and hence the magnetic force never change. Right: they move oppositely; the separation changes by $2a$, so the coupling spring γ adds a force $2\gamma a$ to ka on each magnet.

The relation tested by the experiment. Subtracting the two mode frequencies removes k and m :

$$f_2^2 - f_1^2 = \frac{\gamma}{2\pi^2 m} = \frac{3\mu_0\mu_m^2}{\pi^3 m} d^{-5} \equiv C d^{-5}, \quad \ln(f_2^2 - f_1^2) = \ln C - 5 \ln d. \quad (14)$$

f_1 should be independent of d ; $f_2^2 - f_1^2$ against d on log-log axes should be a straight line of gradient -5 (Figure 4). The intercept gives C and hence μ_m if m is known — or, with μ_m from the data sheet, an absolute test. The size of the coupling is $2\gamma/k = (f_2/f_1)^2 - 1 \approx 0.36$ at the closest separation used (24.5 mm): the magnetic “spring” is about a fifth as stiff as the copper one.

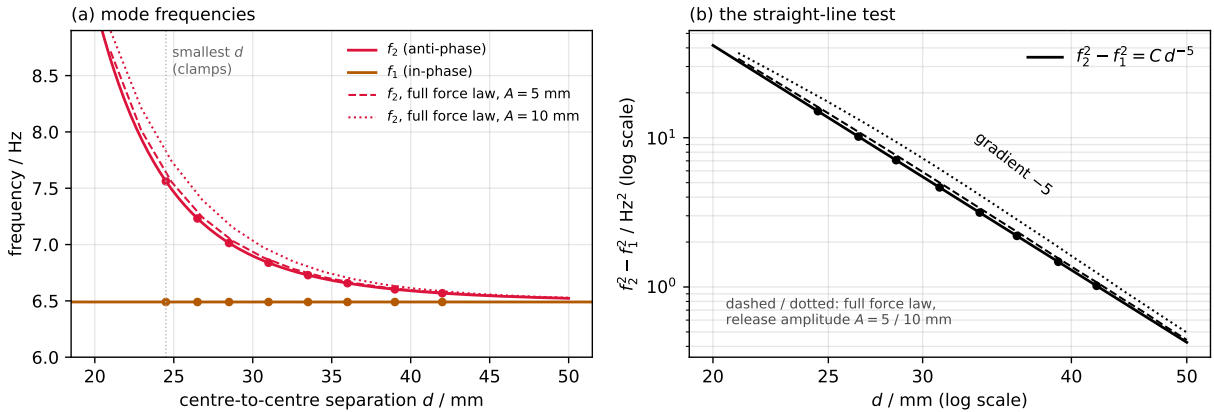


Figure 4: The prediction, drawn with the constants of the pilot run ($f_1 = 6.49$ Hz, $C = 1.33 \times 10^{-7} \text{ m}^5 \text{ s}^{-2}$). Solid: linear model. (a) f_1 is flat; f_2 rises steeply below 30 mm. (b) $f_2^2 - f_1^2$ on logarithmic axes: a line of gradient exactly -5 . Dots: the separations used. Dashed and dotted: the same quantities from the full r^{-4} force law (Section 6) for release amplitudes $A = 5$ and 10 mm — the measured points are expected to lie slightly *above* the line at the smallest d , by an amount that grows with A^2 .

5.5 Damping: the Kelvin–Voigt beam

Copper is not perfectly elastic. The simplest model of internal friction in a solid is the Kelvin–Voigt model: the stress is $\sigma = E(\varepsilon + \tau_v \dot{\varepsilon})$, a spring and a dashpot in parallel, so that the bending

moment acquires a term proportional to the *rate of change of curvature* [4]:

$$\mathcal{M} = EI \left(\frac{\partial^2 y}{\partial s^2} + \tau_v \frac{\partial^3 y}{\partial s^2 \partial t} \right) \implies \mu \frac{\partial^2 y}{\partial t^2} = -EI \frac{\partial^2}{\partial s^2} \left(\frac{\partial^2 y}{\partial s^2} + \tau_v \frac{\partial^3 y}{\partial s^2 \partial t} \right). \quad (15)$$

With the one-mode ansatz $y = x(t)\varphi(s)$ the dissipative term produces, by exactly the same integral as V_{beam} , a force $-k\tau_v \dot{x}$ on the tip coordinate; air drag on the magnet adds a further term $-c_a \dot{x}$. Both are linear in $\dot{x}$ and equal for the two identical springs, so in (11) $m\ddot{x}_i \rightarrow m\ddot{x}_i + c\dot{x}_i$ with $c = k\tau_v + c_a$, and the mode equations become

$$\ddot{\eta} + 2\lambda\dot{\eta} + \omega_1^2\eta = 0, \quad \ddot{\xi} + 2\lambda\dot{\xi} + \omega_2^2\xi = 0, \quad \lambda = \frac{c}{2m} = \frac{1}{\tau_d}, \quad (16)$$

with solutions $e^{-\lambda t} \cos(\omega'_i t + \phi_i)$, $\omega'_i = \sqrt{\omega_i^2 - \lambda^2}$. Three consequences are used in the analysis:

1. **The frequencies are unaffected.** With $\tau_d \approx 21$ s (pilot run) and $\omega \approx 41$ s⁻¹, $\lambda/\omega \approx 10^{-3}$ and $\omega' - \omega \approx -\frac{1}{2}\lambda^2/\omega \sim 10^{-6}\omega$. Damping changes the peak *width* (full width at half maximum of the power spectrum $1/\pi\tau_d \approx 15$ mHz), not the peak position.
2. **Both modes decay at the same rate**, because the same c acts in both. The ratio of the two FFT peaks and the ratio of beat minima to maxima are then constant in time (Figure 5a). A dissipation that depends on the *relative* motion — eddy currents induced in one magnet by the changing field of the other, or air squeezed in the gap — would damp the anti-phase mode faster and make the beat minima rise with time (Figure 5b). The envelope of the record therefore tests which damping mechanism operates.
3. **The KV coefficient is a material property:** $\lambda = k\tau_v/2m + c_a/2m$. Measuring τ_d for the “Infinity” recordings (one spring alone) and at each d separates the two contributions: the KV part is the same for all d .

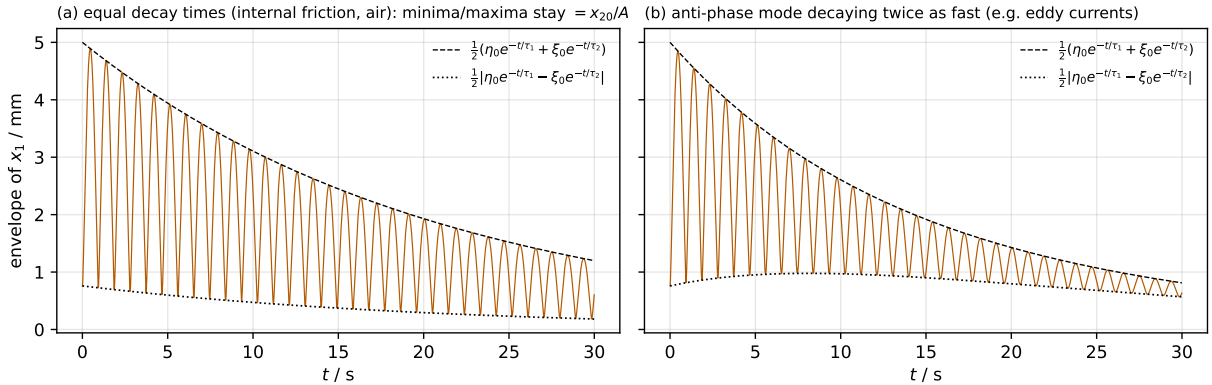


Figure 5: The envelope of the released spring as a test of the damping model ($d = 24.5$ mm, $A = 5$ mm). (a) Equal decay of both modes (Kelvin–Voigt internal friction, air drag on each magnet): the ratio of beat minima to maxima stays fixed at x_{20}/A . (b) If the anti-phase mode decayed twice as fast, the minima would rise while the maxima fall.

5.6 The release experiment: what the spectrum should show

In the experiment spring 1 is pulled outwards by A , held until spring 2 is at rest, and released. While spring 1 is held at $x_1 = -A$ (positive x to the right), spring 2 feels a reduced repulsion and settles a distance $x_{20} = A\gamma/(k + \gamma)$ towards it, at $x_2 = -x_{20}$ (from $kx_2 = \gamma(x_1 - x_2)$, the static form of (11)). Both start from rest, so the mode amplitudes are $\eta_0 = -(A + x_{20})$ and $\xi_0 = A - x_{20}$, and

$$x_1(t) = \frac{1}{2}(\eta_0 \cos \omega_1 t - \xi_0 \cos \omega_2 t)e^{-\lambda t}, \quad x_2(t) = \frac{1}{2}(\eta_0 \cos \omega_1 t + \xi_0 \cos \omega_2 t)e^{-\lambda t}. \quad (17)$$

Each record contains both frequencies; the FFT of *either* spring shows two peaks at f_1 and f_2 whose heights are in the ratio

$$\frac{\xi_0}{|\eta_0|} = \frac{A - x_{20}}{A + x_{20}} = \frac{k}{k + 2\gamma} = \left(\frac{f_1}{f_2}\right)^2. \quad (18)$$

This is a prediction with no free parameter: at $d = 24.5$ mm the anti-phase peak should be 0.74 of the in-phase peak, for both springs. The positions of the peaks do not depend on how the motion was started; the heights do, and (18) is what a hold-and-release start gives. The sum of two cosines is a fast oscillation at $f_{\text{avg}} = \frac{1}{2}(f_1 + f_2)$ whose amplitude swells and shrinks with period

$$T_{\text{beat}} = \frac{1}{f_2 - f_1}, \quad \text{minima } \frac{1}{2}|\eta_0| - \frac{1}{2}\xi_0 = x_{20}, \quad \text{maxima } A, \quad (19)$$

so the depth of the beat minima measures x_{20} and hence γ/k a third way (Figure 6). The energy travels back and forth: spring 2 is loudest when spring 1 is quietest, a quarter of a beat later.

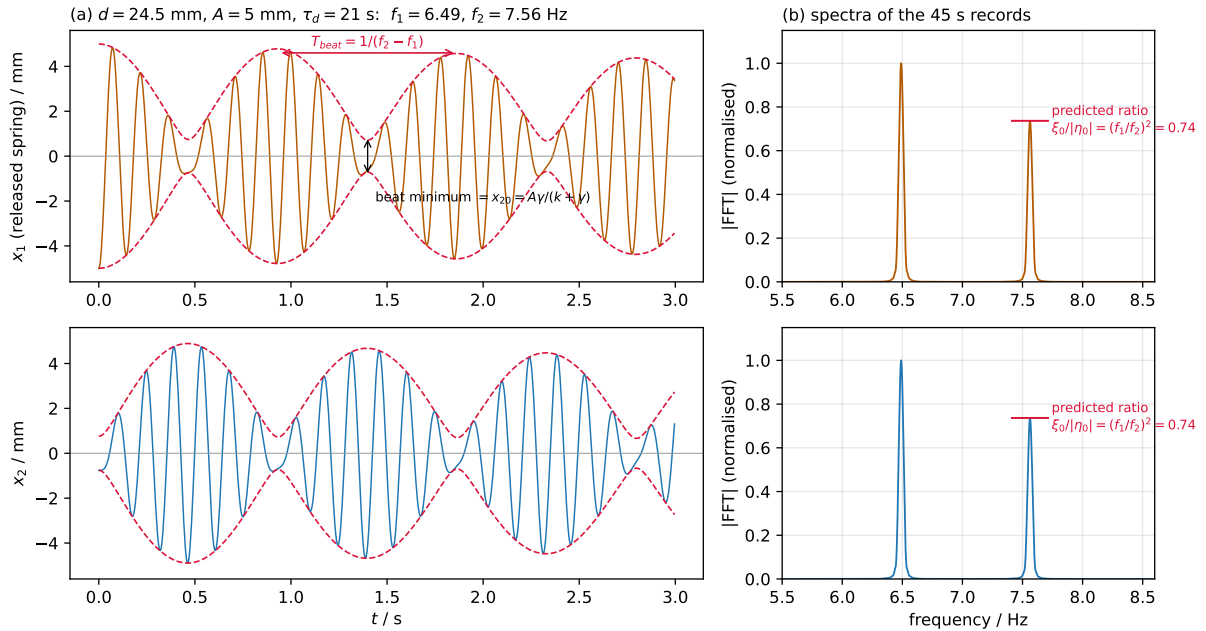


Figure 6: Linear model with Kelvin–Voigt damping, $d = 24.5$ mm, $A = 5$ mm, $\tau_d = 21$ s. (a) Both springs: fast oscillation at f_{avg} , envelope with period $T_{\text{beat}} = 1/(f_2 - f_1) = 0.93$ s, minima at $x_{20} = 0.76$ mm; the envelopes of the two springs are a quarter-beat apart. (b) The spectra of the 45 s records: two peaks, the same positions for both springs, height ratio $(f_1/f_2)^2$.

Summary of the linear model

prediction	equation	how it is tested
f_1 independent of d	(13)	$f_1(d)$ flat; equals the single-spring frequency
$f_2^2 - f_1^2 = Cd^{-5}$	(14)	log–log gradient -5 (main graph)
peak-height ratio $(f_1/f_2)^2$, same for both springs	(18)	FFT peak heights
beat minima = $x_{20} = A\gamma/(k + \gamma)$, period $1/(f_2 - f_1)$	(17)	envelope of $x_1(t)$
equal decay of both modes	§5.5	min/max ratio constant in time
absolute value $C = 3\mu_0\mu_m^2/\pi^3m$	(14)	C from intercept vs data-sheet μ_m , measured m

6 Extension: the non-linear force law

The linearisation replaced $F(d + \xi)$ by its tangent. The true force $F \propto r^{-4}$ is convex: pushing the magnets together by ξ raises the repulsion more than pulling them apart by ξ lowers it. Keeping the full law in the mode equations shows what the linear model misses and when.

Exact mode equations. Adding and subtracting (9) (untilted dipoles) gives, without approximation,

$$m\ddot{\eta} = -k\eta, \quad m\ddot{\xi} = -k\xi + 2[F(d + \xi) - F(d)]. \quad (20)$$

(Each magnet feels $\mp[F(d + \xi) - F(d)]$, the change of the repulsion from its equilibrium value; the difference of the two equations doubles it.) The centre-of-mass motion is *exactly* harmonic at ω_1 : the magnetic force is internal and cancels from η whatever its form. All non-linearity lives in the relative coordinate; for $\xi > 0$ (magnets further apart) the bracket is negative, so the extra force is restoring, and the linear limit $-2\gamma\xi$ is recovered from $F'(d) = -\gamma$. Expanding $F(d + \xi) = F(d)(1 + \xi/d)^{-4}$,

$$(1+u)^{-4} = 1 - 4u + 10u^2 - 20u^3 + \dots, \quad \ddot{\xi} + \omega_2^2 \xi = \omega_2^2 g \left[5\frac{\xi^2}{d} - 10\frac{\xi^3}{d^2} + \dots \right], \quad g \equiv \frac{\gamma}{k + 2\gamma}, \quad (21)$$

where $\gamma = 4F(d)/d$ was used. The parameter g measures how much of the anti-phase stiffness is magnetic ($g = 0.13$ at 24.5 mm, 0.05 at 31 mm, 0.01 at 42 mm), and the expansion parameter is ξ/d .

Perturbation solution. Writing $\xi = a \cos \omega t + \dots$ and collecting terms (Lindstedt–Poincaré, to second order in a/d) gives

$$\xi(t) = \underbrace{\frac{5g}{2} \frac{a^2}{d}}_{\text{mean shift}} + a \cos \omega t - \underbrace{\frac{5g}{6} \frac{a^2}{d} \cos 2\omega t}_{\text{2nd harmonic}} + \underbrace{\left(\frac{5g}{16} + \frac{25g^2}{48} \right) \frac{a^3}{d^2} \cos 3\omega t}_{\text{3rd harmonic}} + \dots \quad (22)$$

$$\frac{\omega - \omega_2}{\omega_2} = \left(\frac{a}{d} \right)^2 \left[\frac{15g}{4} - \frac{125g^2}{12} \right] > 0. \quad (23)$$

Three testable statements follow (all checked against a direct numerical integration of (20), Figure 7):

1. **f_2 depends on amplitude; f_1 never does.** The anti-phase frequency *rises* with the square of the separation swing — the convex force is stiffer on average than its tangent. For the hold-and-release start $a \approx \xi_0 = A - x_{20}$; with $A = 5$ mm the shift is +0.9 % at 24.5 mm, +0.4 % at 31 mm and below 0.1 % beyond 36 mm; with $A = 10$ mm it is four times larger. This is why the release amplitude is a controlled variable, why the measured log–log gradient is expected to bend *above* -5 at the smallest d , and why an amplitude series at fixed d is a direct test of (23). In a damped record the amplitude decays during the measurement, so the FFT peak sits at a time-averaged shift, about 0.15–0.2 of the initial-amplitude value for $\tau_d = 21$ s and $T = 45$ s (numerical result); the two-tone fit over the first few beats sees the full shift.
2. **Harmonics.** The relative motion contains $2f_2$ and $3f_2$ with relative amplitudes $\frac{5g}{6} \frac{\xi_0}{d} \approx 2\%$ and $\approx 0.15\%$ at 24.5 mm, $A = 5$ mm, and a mean outward shift $\frac{5g}{2} \frac{\xi_0^2}{d} \approx 0.2$ mm (the magnets spend more time far apart than close). Since $x_{1,2} = \frac{1}{2}(\eta \mp \xi)$, each spring’s spectrum shows the harmonics of f_2 but not of f_1 ; combination tones $f_1 \pm f_2$, $2f_1$ appear only through the (tiny) tilt coupling of (10). A 2 % line is visible in a 45 s Hann spectrum with a 10^3 dynamic range — an optional check.
3. **Internal resonance.** If $\omega_2 = 2\omega_1$ the second harmonic of the anti-phase motion is resonant with the in-phase mode and energy can pass between modes even at small amplitude — the

“critical value” seen by other groups. Here $\omega_2^2/\omega_1^2 = 1 + 2\gamma/k = 4$ requires $2\gamma/k = 3$, i.e. $d^* = (C/3f_1^2)^{1/5} \approx 16$ mm: below the smallest separation the clamps allow (24.5 mm, where $2\gamma/k = 0.36$) and below the range in which the magnets can be treated as dipoles at all. The condition is therefore not met in this experiment, which is a strength: the two modes stay independent and the linear model remains applicable down to 24.5 mm, with the small amplitude corrections above.

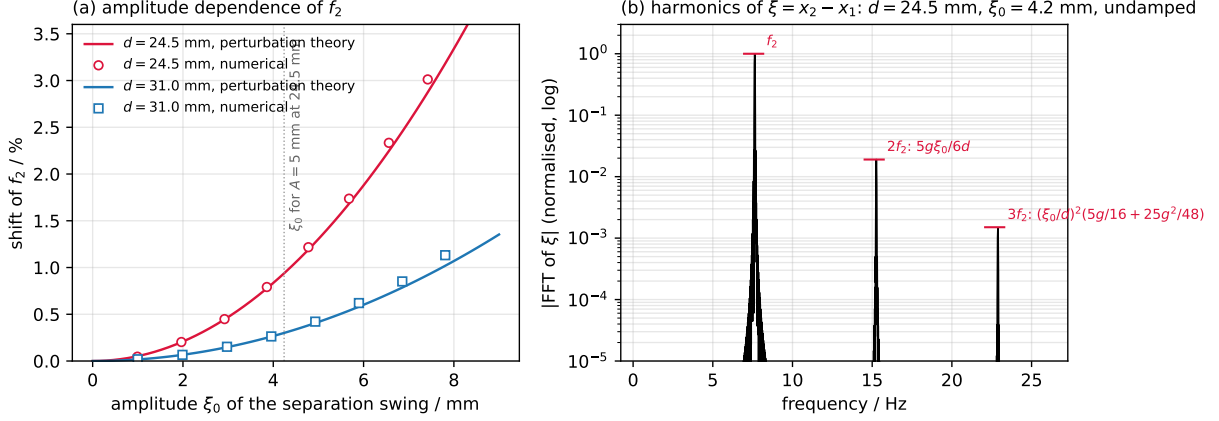


Figure 7: Non-linear predictions checked numerically. (a) Shift of f_2 against the amplitude of the separation swing at two separations: lines, equation (23); symbols, direct integration of (20). The dotted line marks ξ_0 for the standard $A = 5$ mm release at 24.5 mm. (b) Spectrum of $\xi(t)$ from the integration at 24.5 mm (log scale): the harmonics at $2f_2$ and $3f_2$ have the relative sizes predicted by (22) (red bars).

Other neglected effects, in order of size. (i) *Finite magnet size*: a disc magnet is a dipole only for $d \gg D$; the next multipole term changes F by $O(D^2/d^2)$ and steepens the effective exponent at small d — fitted in the evaluation with the magnets modelled as two charged discs. (ii) *Detuning*: if $k_1 \neq k_2$ the modes mix, $(f_2^2 - f_1^2)^2 = (f_b^2 - f_a^2)^2 + (Cd^{-5})^2$, and the peak ratios of the two springs differ; measured by the “Infinity” recordings. (iii) *Tip tilt*: relative corrections $\lesssim 10^{-3}$ to both mode frequencies, equation (10). (iv) *Gravity*: a few per cent of k , absorbed into the measured k . (v) *Higher beam modes*: $\geq 6\omega_1$, not excited by a slow release.

References

- [1] J. M. Gere and B. J. Goodno, *Mechanics of Materials*, 8th ed. (Cengage, 2012), ch. 9 (cantilever deflection) and ch. 5 (bending stress).
- [2] S. S. Rao, *Mechanical Vibrations*, 6th ed. (Pearson, 2017), §2.5 (Rayleigh’s energy method) and §8.5 (cantilever with tip mass).
- [3] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. (Cambridge, 2017), §6.1 (force and energy of magnetic dipoles).
- [4] H. T. Banks and D. J. Inman, “On damping mechanisms in beams”, *J. Appl. Mech.* **58**, 716–723 (1991) (Kelvin–Voigt and air damping of a cantilever).